Chapter 1: Introduction to Research Methods

Why Research Methods Matter

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Overview

Before You Begin

This is your first reading for PS50008A. Take your time with it. The concepts introduced here form the foundation for everything else in this module.

Why Research Methods? A Note for Foundation Year Students

You chose psychology because you want to understand people. Why do we remember some things and forget others? What drives someone to help a stranger—or to hurt them? How do our early experiences shape who we become? These questions drew you here, and they're genuinely worth exploring.

Then you look at your timetable and see "Research Methods and Experimental Design."

If your heart sinks a little, you're not alone - but I'm going to try to turn this around. This material has a reputation, and not a glamorous one. But there are real reasons it's central to your degree—not because we enjoy watching you suffer, but because it changes what you're capable of doing. **It's powerful stuff!**

You're Learning to Ask Questions Properly

Right now, if someone asked you "Does social media cause anxiety?", you might have an opinion. Everyone has one of those. But psychology isn't about opinions—it's about evidence. And evidence doesn't just land in your lap. Someone has to design a way to gather it, collect it carefully, and work out what it actually shows.

That's what you're learning to do. By the end of your degree, you won't just *think* social media might cause anxiety—you'll be able to design a study that could actually test the idea, spot the weaknesses in existing research, and understand why the answer might be more complicated than the headlines suggest.

Think about that for a second... You don't have to rely on anecdote or opinion anymore. You can *create* knowledge.

You'll See Through the Nonsense

We're surrounded by claims about human behaviour. Adverts tell us certain products will make us more confident. Influencers promote supplements that supposedly sharpen your mind. News stories announce that scientists have "proved" all sorts of things about personality, intelligence, or relationships. 9 out of 10 adverts are dubious! (not a real stat!)

Most of these claims are, at best, oversimplified. Many are outright wrong. Understanding research methods gives you the ability to look at a study and ask:

- How many people did they actually test?
- Could something else explain these results?
- Is this finding likely to replicate?

This isn't just academic scepticism. **It's protection against being manipulated** by people who use the language of science without the rigour, without the appropriate 'methods'.

These Skills Travel

Psychology graduates end up in all sorts of places: clinical training, education, marketing, human resources, policy research, user experience design, data analysis, all sorts of entrepreneurial activities I don't understand. What connects these fields is that they all involve understanding people and making decisions based on evidence. It's just thinking scientifically.

The ability to gather data systematically, analyse it honestly, and communicate what it means is valuable almost everywhere. It's also incredibly rare. Many people can express opinions confidently; fewer can actually interrogate those opinions with evidence.

And One More Thing

Some of you will genuinely enjoy this. Not everyone, and probably not immediately, but there's real satisfaction in solving methodological puzzles: working out how to test something that seems untestable, spotting a flaw in an argument, seeing a pattern emerge from data you collected yourself.

I'm not promising you'll love every moment. But we are promising that this material is worth taking seriously—and that taking it seriously might surprise you.

> ! This Week
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> Since Week 11 has no lecture, use this reading to prepare for the term ahead. But above all, **be curious!**

Note

Sophie von Stumm (formerly from Goldsmiths!) and colleagues (2011) sought to identify the best predictors of academic performance. Unsurprisingly perhaps, a person's intelligence was the best predictor. However, they also found that this was matched by the combined predictive power of effort and intellectual curiosity. They conclude that succeeding in academic fields (such as psychology) is not just a question of being smart and trying hard, it is also about having a hungry mind!

Von Stumm, S., Hell, B., & Chamorro-Premuzic, T. (2011). The hungry mind: Intellectual curiosity is the third pillar of academic performance. Perspectives on Psychological Science, 6, 574-588.

Why Do Psychologists Do Research?

Psychology aims to understand human behaviour and mental processes. But how do we move from casual observation to scientific knowledge?

The Problem with Common Sense

We often think we understand human behaviour based on everyday experience - my mum certainly thinks so. But "common sense" has limitations (sorry, mum!):

- **Hindsight bias**: Events seem obvious *after* they occur ("I knew it all along")
- **Confirmation bias**: We notice evidence that supports our beliefs and ignore contradicting evidence
- **Anecdotal evidence**: Individual cases can mislead — your friend's experience isn't representative of everyone

Consider these "common sense" claims:

- "Opposites attract" — But also "Birds of a feather flock together"
- "Too many cooks spoil the broth" — But also "Many hands make light work"
- "Absence makes the heart grow fonder" — But also "Out of sight, out of mind"

Common sense can justify almost any conclusion. Research methods provide systematic tools to overcome these biases and determine which claims actually hold up under scrutiny.

Why Intuitions About Behaviour Are Often Wrong

Our intuitions fail us for several reasons:

1. **We don't see our own blind spots** — We're unaware of the biases affecting our judgement
2. **Memory is reconstructive** — We misremember events to fit our current beliefs
3. **We overgeneralise from small samples** — One or two examples feel like proof
4. **We confuse correlation with causation** — Two things happening together doesn't mean one caused the other

For example, studies consistently show that:

- People are poor at predicting what will make them happy
- Eyewitness testimony is far less reliable than we assume
- "Venting" anger often increases rather than decreases aggression

These findings contradict common intuitions — which is precisely why we need systematic research.

What Makes a Claim Scientific?

The Difference Between Opinion and Science

An **opinion** is a personal belief that may or may not be supported by evidence: > "I think social media makes people unhappy."

A **scientific claim** is testable, based on systematic evidence, and open to being proven wrong: > "In a study of 1,000 adults, those who used social media for more than 3 hours daily reported lower life satisfaction scores ($M = 5.2$) than those who used it less than 1 hour ($M = 6.8$), $t(998) = 4.32$, $p < .001$."

Four Hallmarks of Scientific Claims

1. **Empirical basis** — Based on observable, measurable evidence, not just reasoning or authority
2. **Systematic collection** — Data gathered using consistent, documented methods
3. **Falsifiability** — The claim could, in principle, be proven wrong by evidence
4. **Transparency** — Methods are described clearly enough that others could repeat them

What "Replicable" Means

A finding is **replicable** if other researchers, following the same procedures, get the same results.

Why does this matter?

- A single study might have flaws we don't notice
- Random chance can produce misleading results
- Researcher bias might influence findings

When multiple independent teams find the same result, we can be more confident it's real.

> **i The Replication Crisis**
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> Psychology has faced a "replication crisis" — many classic findings failed to replicate when researchers tried to repeat them. This has led to better practices: larger samples, pre-registration of hypotheses, and more transparency. You'll learn about this in Week 17.

The Scientific Method

Researchers typically follow these steps:

1. Identify a Question

Start with curiosity. What do you want to understand?

> "Do people remember information better if they test themselves rather than just re-reading?"

2. Review Existing Knowledge

What have others already discovered? Build on previous work rather than starting from scratch.

3. Formulate a Hypothesis

Make a specific, testable prediction.

> "Students who use self-testing will recall more information after one week than students who re-read the material."

4. Design a Study

Plan exactly how you'll test your hypothesis: - Who will you study? - What will you measure? - How will you control for other explanations?

5. Collect Data

Gather observations systematically, following your planned procedures.

6. Analyse Results

Use statistics to determine whether your findings are meaningful or could have occurred by chance.

7. Draw Conclusions

What do your results mean? Do they support your hypothesis? What are the limitations?

8. Report Findings

Share your methods and results so others can evaluate and build on your work.

You'll work through all these steps in your group project this term.

Variables: Things That Vary

A **variable** is anything that can take different values.

Examples of Variables

Variable	Possible Values
Height	150cm, 165cm, 180cm…
Reaction time	200ms, 350ms, 420ms…
Mood	Happy, sad, anxious, neutral…
Number of words recalled	0, 1, 2, 3… 20
Exam score	0-100

Why Variables Matter

Research is fundamentally about understanding how variables relate to each other:

- Does **study time** affect **exam performance**?
- Does **sleep quality** predict **reaction time**?
- Is **age** related to **memory capacity**?

In DataLab, you'll measure several variables and see how they differ between people and across conditions.

Types of Variables

Continuous variables can take any value within a range: - Height (175.3cm) - Reaction time (287.4ms) - Temperature (36.8°C)

Categorical variables fall into distinct categories: - Gender (male, female, non-binary) - Experimental condition (control, treatment) - Handedness (left, right, ambidextrous)

Discrete variables are countable whole numbers: - Number of items recalled (7) - Number of errors (3) - Number of children (2)

Understanding variable types helps you choose appropriate statistical analyses — something you'll learn throughout this module.

How Empirical Evidence Helps Us Understand Behaviour

Moving Beyond Speculation

Without data, debates about human behaviour often go in circles. Everyone has opinions, but how do we know whose is correct?

Empirical evidence allows us to:

1. **Test competing theories** — Instead of arguing, we can design studies to see which prediction holds
2. **Discover unexpected findings** — Data often reveals patterns we wouldn't have predicted
3. **Quantify effects** — Not just "X affects Y" but "X increases Y by approximately this much"

4. **Identify boundary conditions** — Under what circumstances does a finding hold or not hold?

An Example: Does Punishment Deter Crime?

Common sense might suggest that harsher punishments reduce crime. But what does the evidence show?

Research has found: - **Certainty** of punishment matters more than **severity** - Very harsh sentences may actually increase recidivism in some cases - Deterrent effects vary by crime type and offender characteristics

Without systematic data collection, we'd have no way to discover these nuances.

Connecting to DataLab

Your Data, Your Learning

In Thursday's lab, you'll experience research from both perspectives:

- As a **participant** — completing cognitive tasks
- As a **tester** — administering tasks and recording data

This dual perspective helps you understand:

- Why standardised procedures matter (you'll see what happens when instructions vary)
- How measurement introduces variability (the same person gives different responses across trials)
- What it feels like to generate data that will be analysed

The data you generate will be used throughout the term. When we discuss distributions in Week 12, we'll look at the distribution of YOUR reaction times. When we learn about t-tests in Week 14, we'll compare performance between Room A and Room B. This is your data, and you'll learn statistics by exploring it.

Summary

Concept	Key Point
Common sense	Unreliable — biased by hindsight, confirmation, and anecdote
Scientific claims	Testable, based on systematic evidence, falsifiable
Replicability	Findings should hold when others repeat the study
Variables	Anything that can take different values
Empirical evidence	Allows us to test theories rather than just argue

Check Your Understanding

Before the lab, make sure you can answer these questions:

1. Why can't we rely on common sense to understand human behaviour?
2. What makes a claim "scientific" rather than just an opinion?
3. What does it mean for a finding to be "replicable" and why does this matter?
4. What is a variable? Give three examples.
5. Why do researchers follow systematic methods rather than just observing casually?

Further Reading

If you want to explore these ideas further:

- Coolican, H. (2017). Chapter 1: Psychology and research. VLeBooks
- Harris, P. (2008). Chapter 1: Introduction to psychological research. VLeBooks
- Seeing Theory: Basic Probability — Interactive introduction to randomness and variability